



Use Case Scope & Diagrams

Last Revision: 29 September 2026

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TABLE OF CONTENTS

[Acronyms Used..... 2](#)

[Subsystem 1: Authentication..... 2](#)

[Subsystem 2: Ingestion..... 5](#)

[Subsystem 3: Risk..... 6](#)

[Subsystem 4: Field Reports and Routes..... 8](#)

[Subsystem 5: Tip-off..... 10](#)

[Subsystem 6: Dashboard..... 11](#)

[Subsystem 7: Admin..... 12](#)

[Subsystem 8: Geospatial..... 13](#)

Acronyms Used

- **TUCBW** - This Use Case Begins With
- **TUCEW** - This Use Case Ends With

Subsystem 1: Authentication

NOTE: This subsystem is included for completeness, even if they do not count towards the total use case count

Image

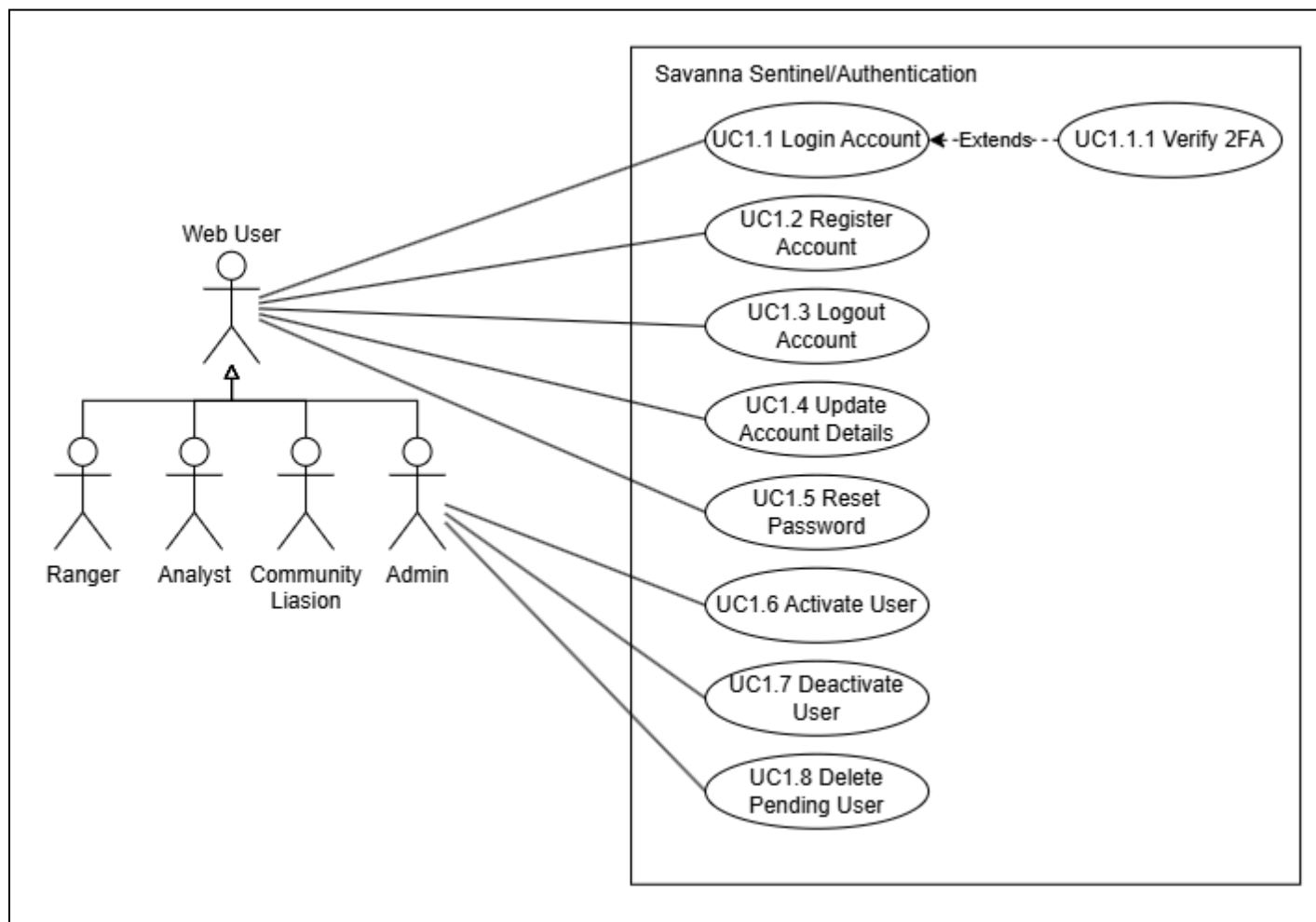


Figure 1 - Authentication Subsystem

Use Case Scope

Use Case Number	Starts With/Ends With
UC1.1 Login Account	• TUCBW the user is shown the login screen upon entering the website. • TUCEW the user being shown a confirmation message and being redirected to the dashboard. • UC 1.1.1 If the user is an admin, extend to Verify 2FA ○ TUCBW the user being prompted to enter a code to continue ○ TUCEW the user entered the code and continuing with UC1.1
UC1.2 Register Account	• TUCBW the user clicking the "Register Account" button on the login screen. • TUCEW the user being shown a confirmation message and being told to initiate UC1.1.
UC 1.3 Logout Account	• TUCBW the user clicking the Logout button on the navigation burger menu. • TUCEW the user being shown the login screen.
UC 1.4 Update Account Details	• TUCBW the user entering the submission details and clicking the submit button • TUCEW the user being informed their details have been updated. • ALT TUCEW the user being redirected to the login screen, only if their password has changed.
UC 1.5 Reset password	• TUCBW the user clicking the reset password button on the login screen • TUCEW the user entering their new password on the magic link and being redirected to login
UC 1.6 Activate User	• TUCBW the admin clicking the activate button next to the corresponding pending user • TUCEW the admin being shown a confirmation message, and the user

	being sent a welcome email.
UC 1.7 Deactivate User	• TUCBW the admin clicking the deactivate button next to the corresponding active user • TUCEW the admin being shown a confirmation message, and the user being sent a deactivation email. Important to note the account is not removed from the DB, just access to the service is revoked.
UC 1.8 Delete Pending User	• TUCBW the admin clicking the reject button next to the corresponding pending user. • TUCEW with the admin being shown a confirmation message, and the user being sent a rejection email.

Subsystem 2: Ingestion

Image

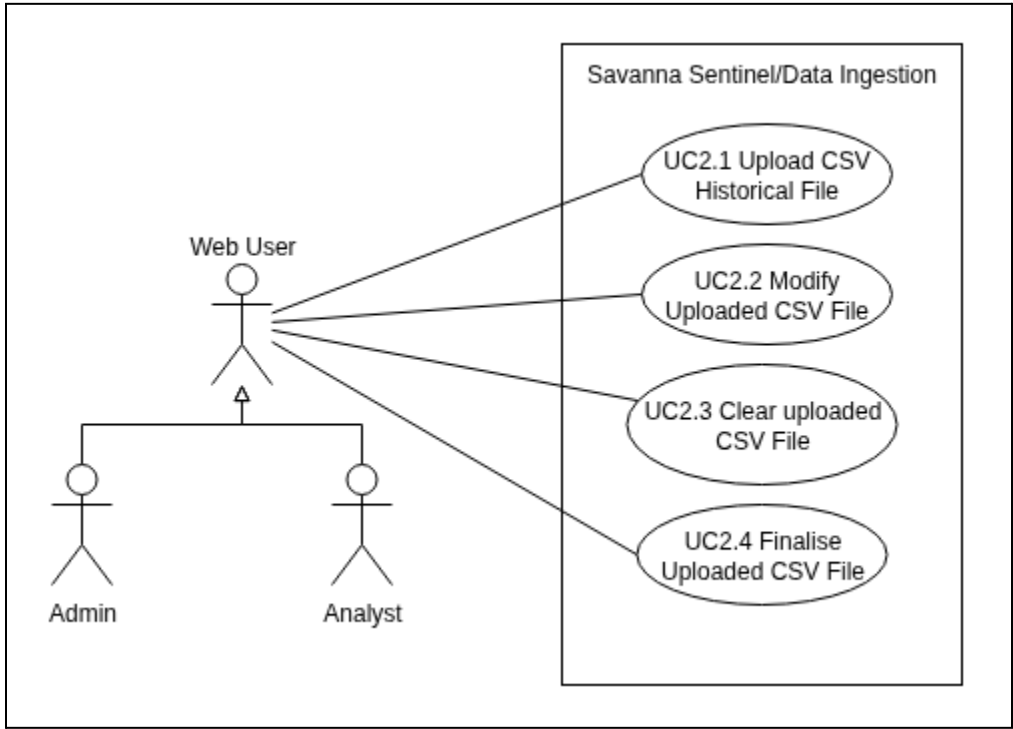


Figure 2 - Ingestion Subsystem

Use Case Scope

Use Case Number	Starts With/Ends With
UC2.1 Upload CSV Historical File	• TUCBW the user clicking the upload file button and choosing a csv file to upload. • TUCEW the system displaying the parsed CSV file and any errors that it could not remedy.

UC2.2 Modify Uploaded CSV File	• TUCBW the user selecting a malformed record from the uploaded file. • TUCEW the system confirming the new details are parseable and in a valid format.
UC2.3 Clear Uploaded CSV File	• TUCBW the user clicking the clear button, after uploading a CSV file. • TUCEW the user being redirected to the beginning of UC2.1.
UC2.4 Finalise Uploaded CSV File	• TUCBW the user clicking the Upload button. • TUCEW the system confirming the file has been uploaded to the database to be analysed by the risk engine, and the user being redirected to the beginning of UC2.1.

Subsystem 3: Risk

Image

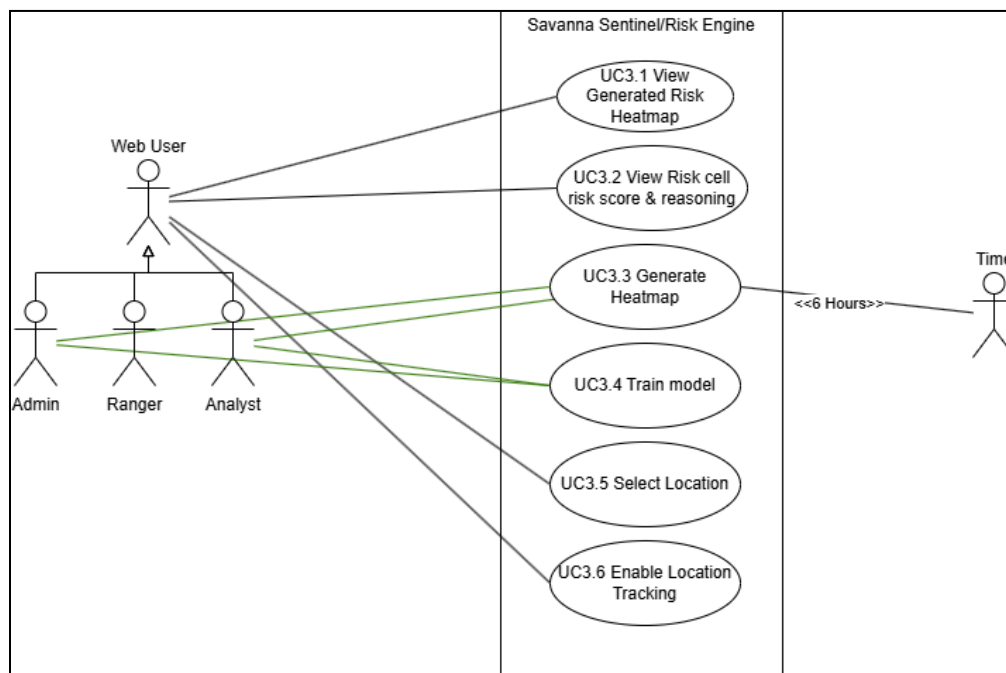


Figure 3 - Risk Subsystem

Use Case Scope

Use Case Number	Starts With/Ends With
UC3.1 View Generated Risk Heatmap	• TUCBW the user entering the map screen. • TUCEW the user being presented the most recent heatmap their role permits generated by the AI risk engine
UC 3.2 View Risk cell risk score & reasoning	• TUCBW the user clicking a cell on the map. • TUCEW with the user being presented information generated by the AI risk engine on the selected cell.
UC 3.3 Generate Heatmap	• TUCBW the user clicking the generate heatmap button, or when 6 hours have elapsed. • TUCEW a new heatmap being generated with parameters found in the system and being displayed to the user.
UC 3.4 Train Model	• TUCBW the user clicking the train model button. • TUCEW a new ai model being trained to read the parameters on the system.
UC 3.5 Select Location	• TUCBW the user enabling location and the application being unable to determine a previously known location of the user. • TUCEW the user selecting a POI and their location being set to match that POI
UC 3.6 Enable Location Tracking	• TUCBW the user clicking the enable location button • TUCEW the users location being displayed on the map.

Subsystem 4: Field Reports and Routes

Image

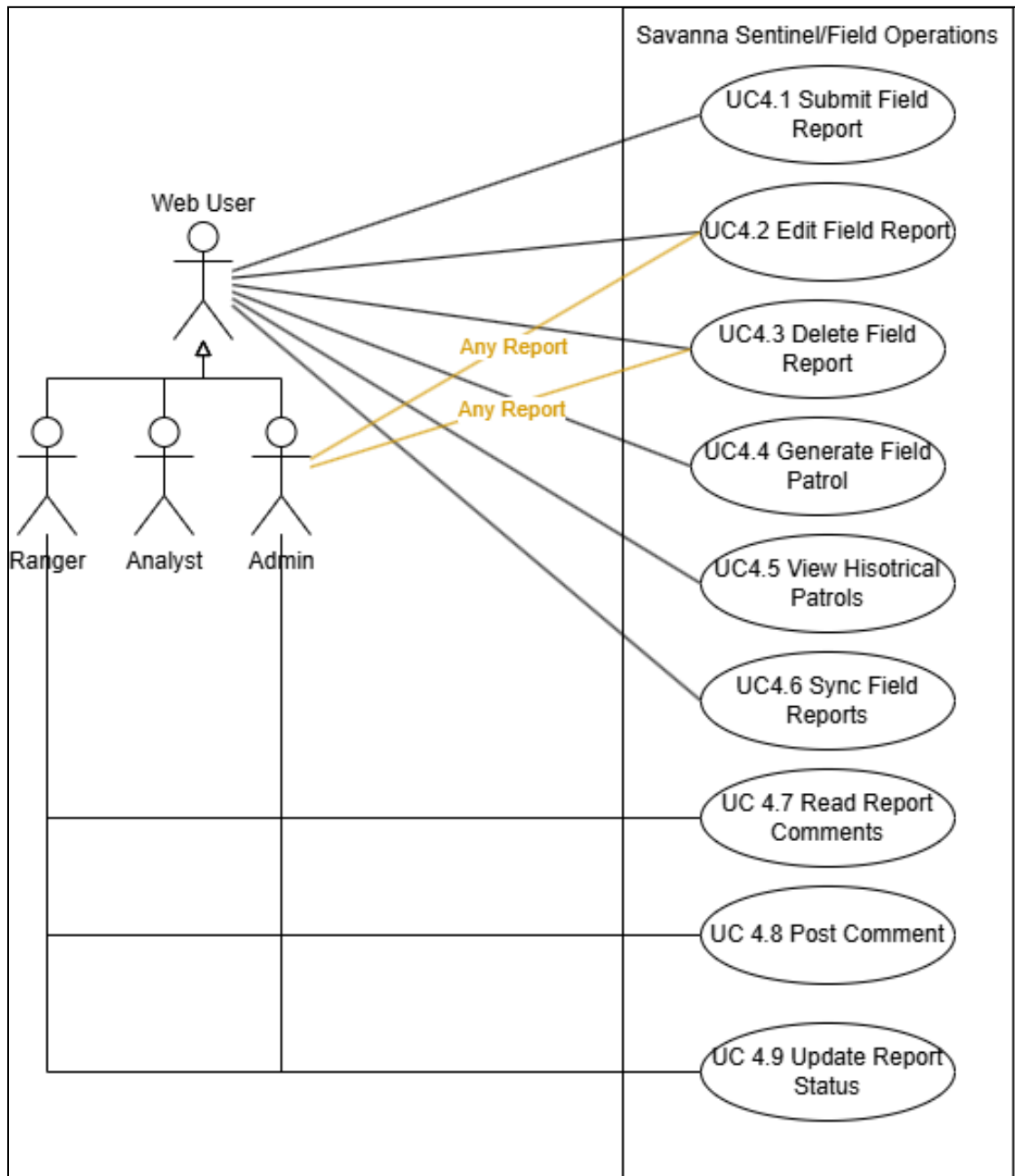


Figure 4 - Field Report Subsystem

Use Case Scope

Use Case Number	Starts With/End With
UC4.1 Submit Field Report	• TUCBW the user clicking the submit report button. • TUCEW the field report being submitted to the system or being queued for upload
UC4.2 Edit Field Report	• TUCBW the user selecting a pending, unuploaded, submitted field report. • TUCEW the pending field report being updated and the user being notified.
UC4.3 Delete Field Report	• TUCBW the user selecting a pending, unuploaded, submitted field report and clicking the delete button. • TUCEW the user being notified the field report is no longer queued for upload.
UC4.4 Generate Field Patrol	• TUCBW the user clicking the generate patrol on the risk heatmap. • TUCEW the user being shown potential routes for the patrol.
UC4.5 View Historical Patrols	• TUCBW the user clicking the show previous patrol routes button. • TUCEW the user being shown their previous generated patrol routes.
UC4.6 Sync Field Reports	• TUCBW automatically, or the user pressing the sync button. • TUCEW the user being notified that all pending field reports have been uploaded.
UC 4.7 Read Report comments	• TUCBW the user clicking a submitted report • TUCEW the user seeing a comment thread of that specific report
UC 4.8 Post comment	• TUCBW the user typing their comment and uploading photos • TUCEW the comment being visible to all other authorised users and the user seeing their own comment

	posted
UC 4.9 Update Report Status	• TUCBW the user selecting a new status for the report • TUCEW the user being shown the report status has been updated in the comment thread.

Subsystem 5: Tip-off

Image

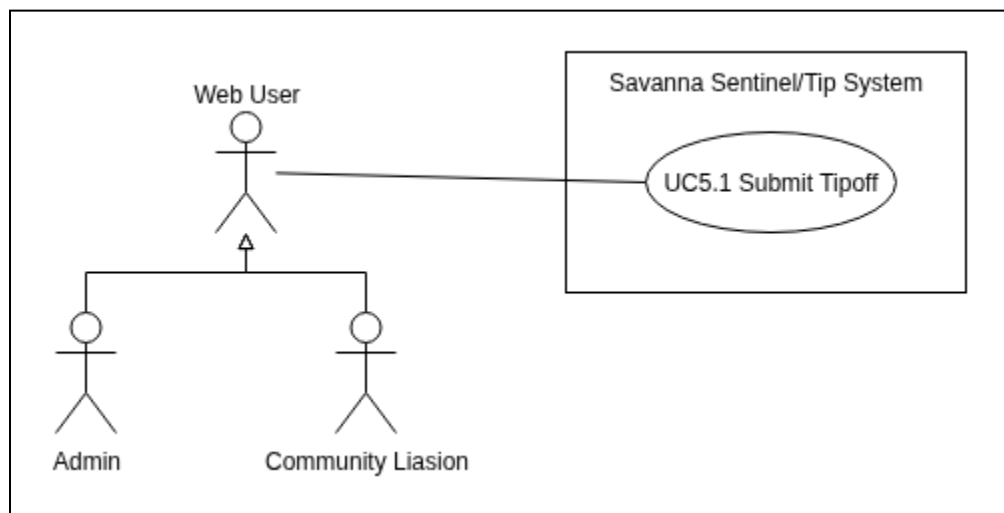


Figure 5 - Tip-off Subsystem

Use Case Scope

Use Case Number	Starts With/End With
UC5.1 Submit Tipoff	• TUCBW the user entering tip off details and clicks the submit button. • TUCEW the user being informed that a tip off has been made, and that it will be processed by the risk engine.

Subsystem 6: Dashboard

Image

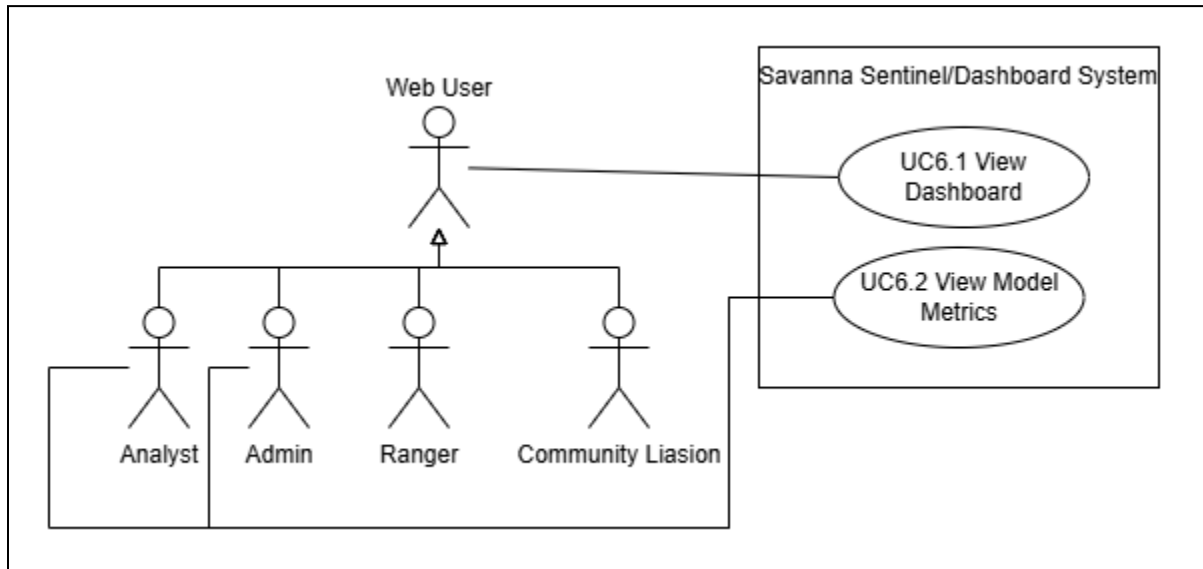


Figure 6 - Dashboard Subsystem

Use Case Scope

Use Case Number	Starts With/End With
UC6.1 View Dashboard	• TUCBW the user viewing the dashboard page. • TUCEW the user being presented their respective role's dashboard.
UC6.2 View Model Metrics	• TUCBW with the user expanding the model section on the dashboard. • TUCEW the user being shown the risk engines performance.

Subsystem 7: Admin

Image

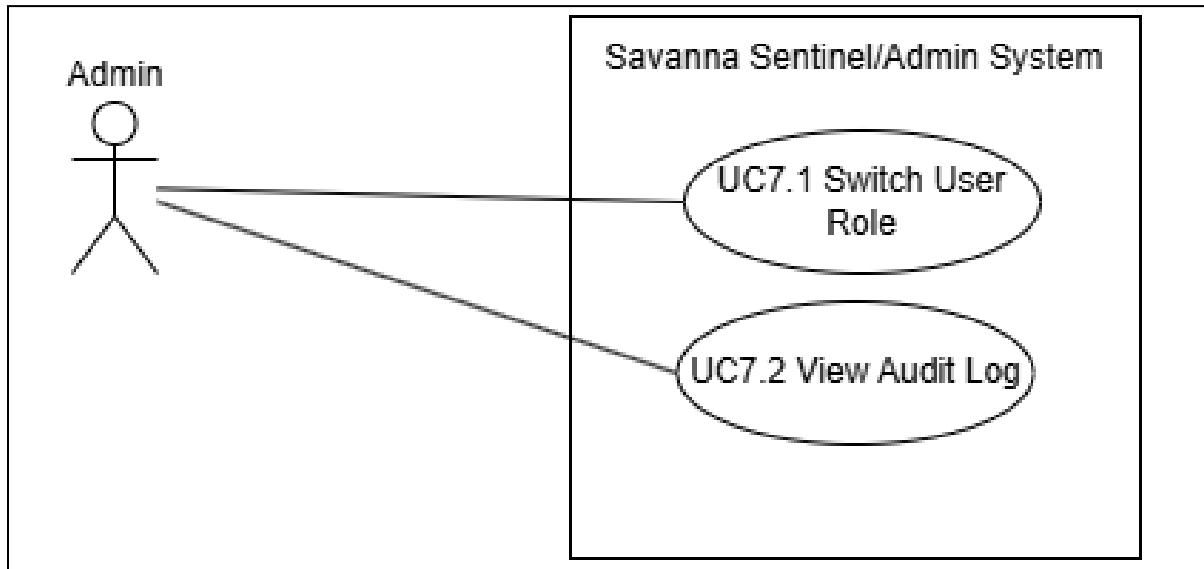


Figure 7 - Admin Subsystem

Use Case Scope

Use Case Number	Starts With/End With
UC7.1 Switch User Role	• TUCBW the admin selecting a role to switch a user to. • TUCEW the admin being shown a message that the operation was successful
UC6.2 View Model Metrics	• TUCBW the admin clicking the audit log button on the admin page • TUCEW the admin being shown the audit log in a tabular format.

Subsystem 8: Geospatial

Image

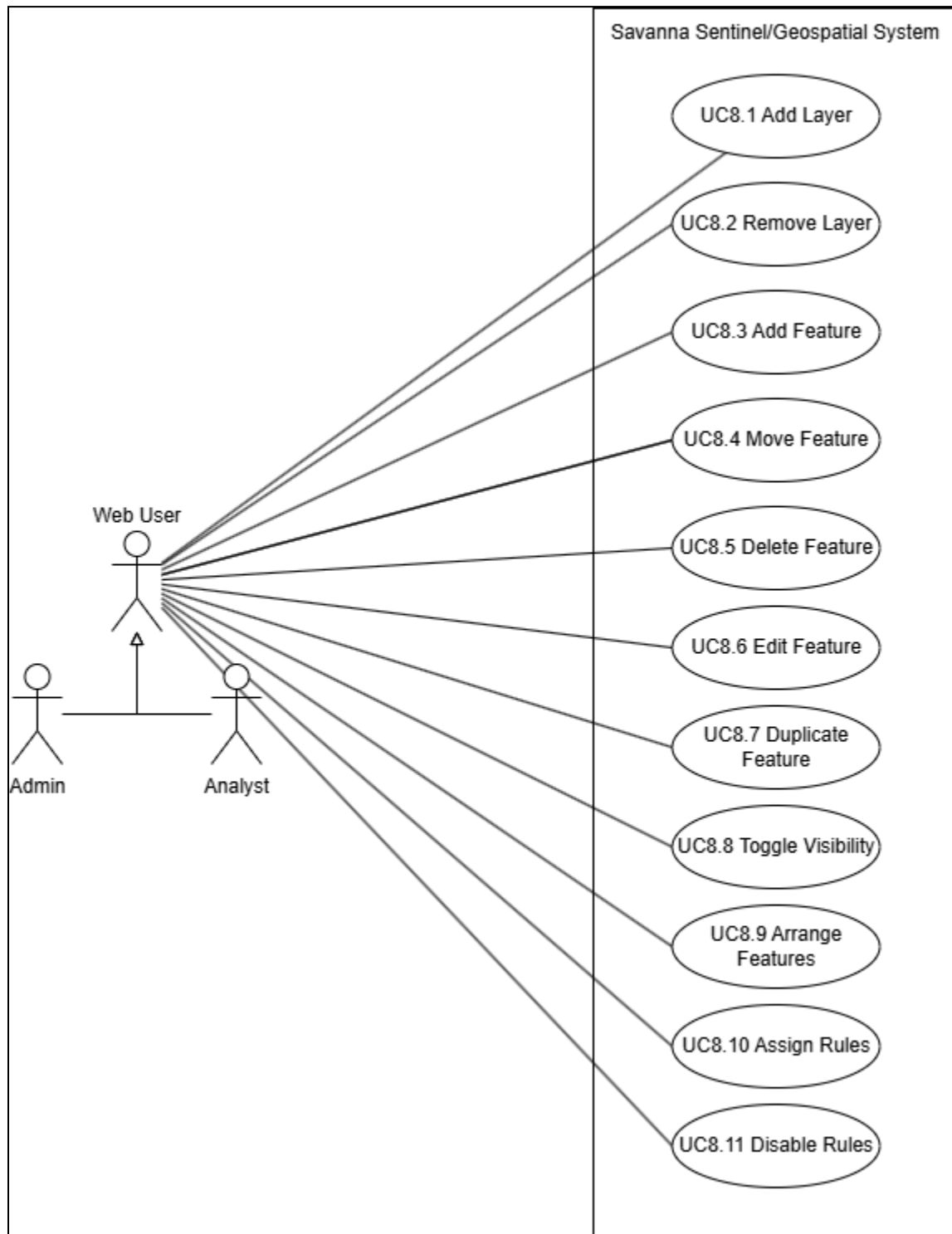


Figure 8 - Geospatial Subsystem

Use Case Scope

Use Case Number	Starts With/End With
UC 8.1 Add Layer	• TUCBW the admin selecting a role to switch a user to. • TUCEW the admin being shown a message that the operation was successful
UC 8.2 Remove Layer	• TUCBW the admin clicking the audit log button on the admin page • TUCEW the admin being shown the audit log in a tabular format.
UC 8.3 Add Feature	• TUCBW the user selecting a layer and then one of the drawing tools. • TUCEW an element being added to the selected layer.
UC 8.4 Move Feature	• TUCBW the user selecting a feature and clicking more actions and move feature • TUCEW the feature being moved to another layer
UC 8.5 Delete Feature	• TUCBW the user selecting a feature and clicking more actions and remove feature. • TUCEW the feature being removed from the map.
UC 8.6 Edit Feature	• TUCBW the user selecting a feature. • TUCEW the feature being updated with the specified details on the edit panel.
UC 8.7 Duplicate feature	• TUCBW the user selecting a feature and clicking more actions and duplicate feature. • TUCEW the feature and all of its properties being duplicated.
UC 8.8 Toggle visibility	• TUCBW the user clicking the box next to a listed feature. • TUCEW the feature being visually removed from the map.

UC 8.9 Arrange Feature	• TUCBW the user dragging a feature. • TUCEW the feature being moved to a different position in the hierarchy, changing its rendering priority.
UC 8.10 Assign Rules	• TUCBW the user selecting a feature and clicking edit rules. • TUCEW the feature's rules and its impact on the heatmap being edited.
UC 8.11 Disable Rules	• TUCBW the user selecting a feature and clicking the disable rules button. • TUCEW the feature's impact on the heatmap being disabled, but still visually displaying.